Workout 1B Homework 2 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:16

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

In the sequence which starts with the terms $1, 3, 6, 10, 15, 21, \ldots,$ the nth term is equal to what?

- The sum of the first n cubes
- The sum of the first n even numbers
- The sum of the first n odd numbers
- The sum of the numbers from 1 to n
- The sum of the first n squares

2

Review Later

Suppose we have a set with elements $\{0, -1, -2, -2, -3, -3, -3, 4, 4, 4, 4, 5, 5, 5, 6, 6, 7\}$. Which of the following will change if we replace each element x in the set with $|x|$? Choose all that apply.

- [ ] Mean
- [ ] Median
- [ ] Mode
- [ ] Range

3

Review Later

Which of the following is closest to the value of $\log_{10} \pi$?

- $\frac{1}{2}$
- $\frac{2}{3}$

$\frac{1}{7}$

2

4

Review Later

What is the biggest prime factor of $(22,206)^2 - (5,822)^2$?

5

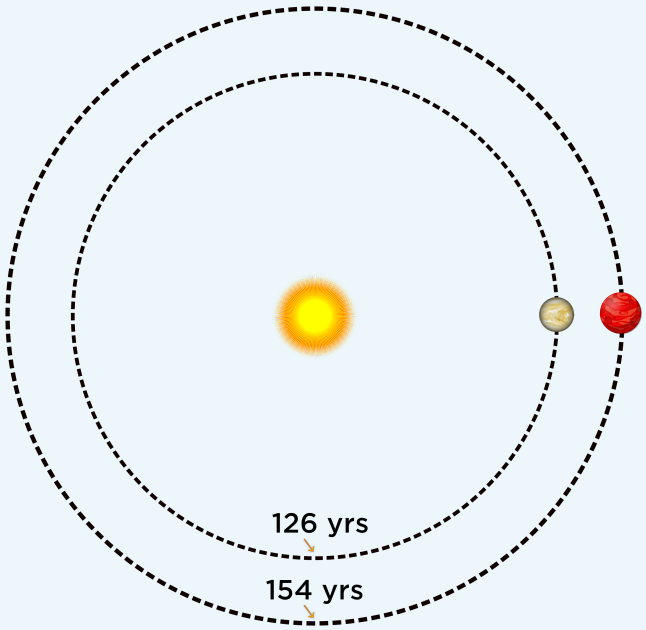
Review Later

What is the 1.2^{th} root of 64?

6

Review Later

Two planets are orbiting a star, with orbit lengths of 154 years and 126 years. Right now, the planets are lined up to the right of the star. After how many years will the two planets be lined up at this position again?



7

Review Later

Alan lives on the odd-numbered side of his street. Let a equal the number of houses to the left, in the increasing direction, next to Alan's house, and let x equal the number of Alan's house. In terms of a and x, what is the sum of the house numbers to the left of Alan's house, not including Alan's house?

- $2x + 2 + 2a$
- $\frac{b(b+1)+a(a+1)}{a-b}$
- $a(x + 1 + a)$
- $2(1 + a)a$
- $2(x + 1 + a)a$
- $ax + 2a$

8

Review Later

Elena is standing in the middle between two houses, on one side of a street full of houses. The house numbers on any side are either all even or all odd. If she totals the numbers of the houses on either side of her, the total is the same. What is the smallest possible average of the two house numbers on either side of Elena?

9

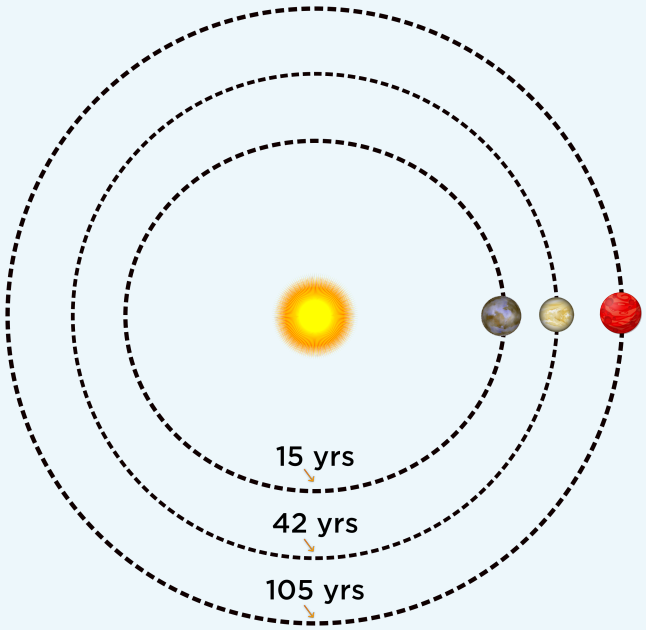
Review Later

What number comes next in this sequence: $3, 8, 24, 48, 120, 168, ?$

10

Review Later

Three planets are orbiting a star, with orbit lengths of 105 years, 42 years, and 15 years. Right now, the planets are all lined up in a perfect eclipse. After how many years will the planets and star be collinear again, in the sense that there is a straight line that goes through all of them?



Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

Suppose there is a three-digit number abc such that the number equals $a + b^2 + c^9 - c^2 - c$. Which of the following is a possible value for c?

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

12

Review Later

Which of the following is equal to $\frac{1000}{996}$?

- $1 + \frac{4}{1000}\left(1 + \frac{4}{1000}\right)$
- $1 + \frac{4}{1000}\left(1 + \frac{4}{996}\right)$
- $1 + \frac{4}{1000}\left(1 - \frac{4}{996}\right)$
- $1 + \frac{4}{996}\left(1 + \frac{4}{1000}\right)$
- $1 - \frac{4}{1000}$

13

Review Later

Using only a single 1 and a single 2, and any operation including $+, -, \times, \div$, along with using powers, factorials or decimals, what is the maximum value you can get? (Note: adding a 0 isn't allowed.)

- 2
- 3
- 5
- 10
- 20
- 100
- 200

14

Review Later

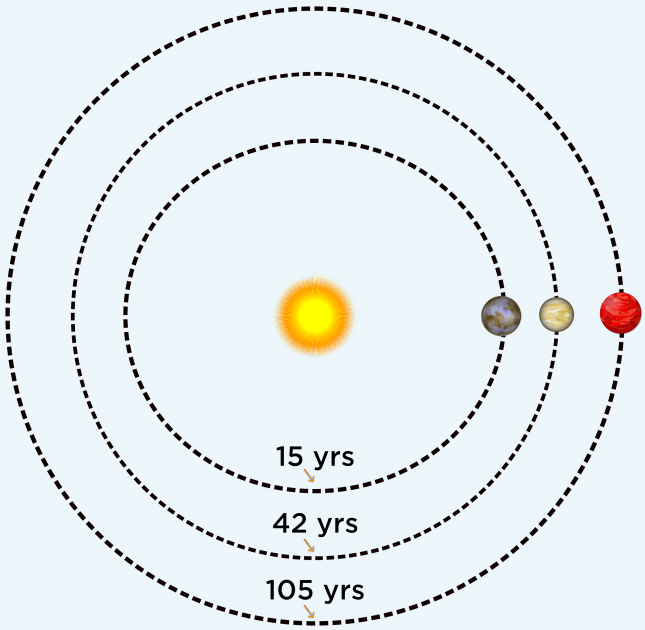
What is the value of

$$\log_{10}\left[3(1^2)+3(2^2)+3(3^2)+3(4^2)+3(5^2)+3(6^2)+3(7^2)+3(8^2)+3(9^2)+3(10^2)\right.$$
$$\left.-1-3-5-7-9-11-13-15-17-19-55\right]?$$

15

Review Later

Three planets are orbiting a star, with orbit lengths of 105 years, 42 years, and 15 years. Right now, the planets are all lined up in a perfect eclipse. After t years, the planets and star will be collinear again, in the sense that there is a straight line that goes through all of them. How many orbits will the slowest planet and fastest planet have gone after time t? Your answers should be fractions in the form $\frac{a}{b}$, where a and b are integers with no common factors.



Slowest planet: _____ / _____ orbits

Fastest planet: _____ / _____ orbits

16

Review Later

What number comes next in the sequence $1, 4, 3, 11, 15, 13?$

17

Review Later

Convert the fraction $\frac{10}{9996}$ into a decimal. Find the sum of the first five *non-zero* digits of this decimal. (It's not zero!)

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

Alan lives on the even-numbered side of his street. If he totals the numbers of the houses on either side of him, the total is the same. If Alan's house number is a two-digit multiple of 10, what is the smallest possible value of Alan's house?

- 10
- 20
- 30
- 40
- 50
- 60

19

Review Later

Suppose there is a three-digit number abc such that the number equals $a + b^2 + c^9 - c^2 - c$. What is the sum of the possible three-digit numbers?

20

Review Later

What number comes next in this sequence: $1, 2, 2, 3, 2, 4, 2, 4, 3, 4, 2, 6, 2, 4, 4, ?$

View Marked Questions

Finished

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